

simplifynext

ignite
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AGENTIC AI HACKATHON 2026

 Fueling Tomorrow's Changemakers



Centre for
Future-ready Graduates



Welcome & Vision



'ignite' is SimplifyNext's initiative to inspire and empower the next generation of changemakers.

Shaping the Future of Work Through Agentic AI Innovation

This Hackathon is your launchpad to learn, experiment, and build with the next frontier of AI: Agentic AI



Why Agentic AI Matters

Agentic AI is the next generation of AI — systems that don't just respond, but think, adapt, and act to achieve goals.

From Tools to Agents



Old AI responds to prompts. Agentic AI can reason, plan, adapt, and take action to achieve goals.

The AI Shift is Here



Agentic AI is accelerating how we automate, innovate, and solve problems.

Future-Proofing Your Career



Singapore is training 40,000 tech professionals in AI by 2029 — start building today.

What's In It for You



Build Real AI Agents
(not just chatbots)



Win Awesome Prizes
(cash, gadgets, swag, and recognition)



Hands-on project experience
(less theory, more building)



Resume-boosting portfolio project
(stand out to recruiters)



Mentorship from industry experts
(real experts, real guidance)



Potential internship or job
(network with industry leaders)

Problem Statement

Design for a World in Transformation

Change is everywhere – in how we live, learn, and relate to one another. Transformation takes time, effort, and the right support at the right moment.

This is your chance to build something that helps. We envision a solution that plans, acts, and adapts over time.

Your team will choose the problem and decide who it serves. You will design a solution that thinks ahead, takes action, and leaves people genuinely better off.

Meet the Speaker

Joon Sern Lee

Chief Data Scientist
SimplifyNext

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How AI is Used in Industry Today

1. Automating knowledge work

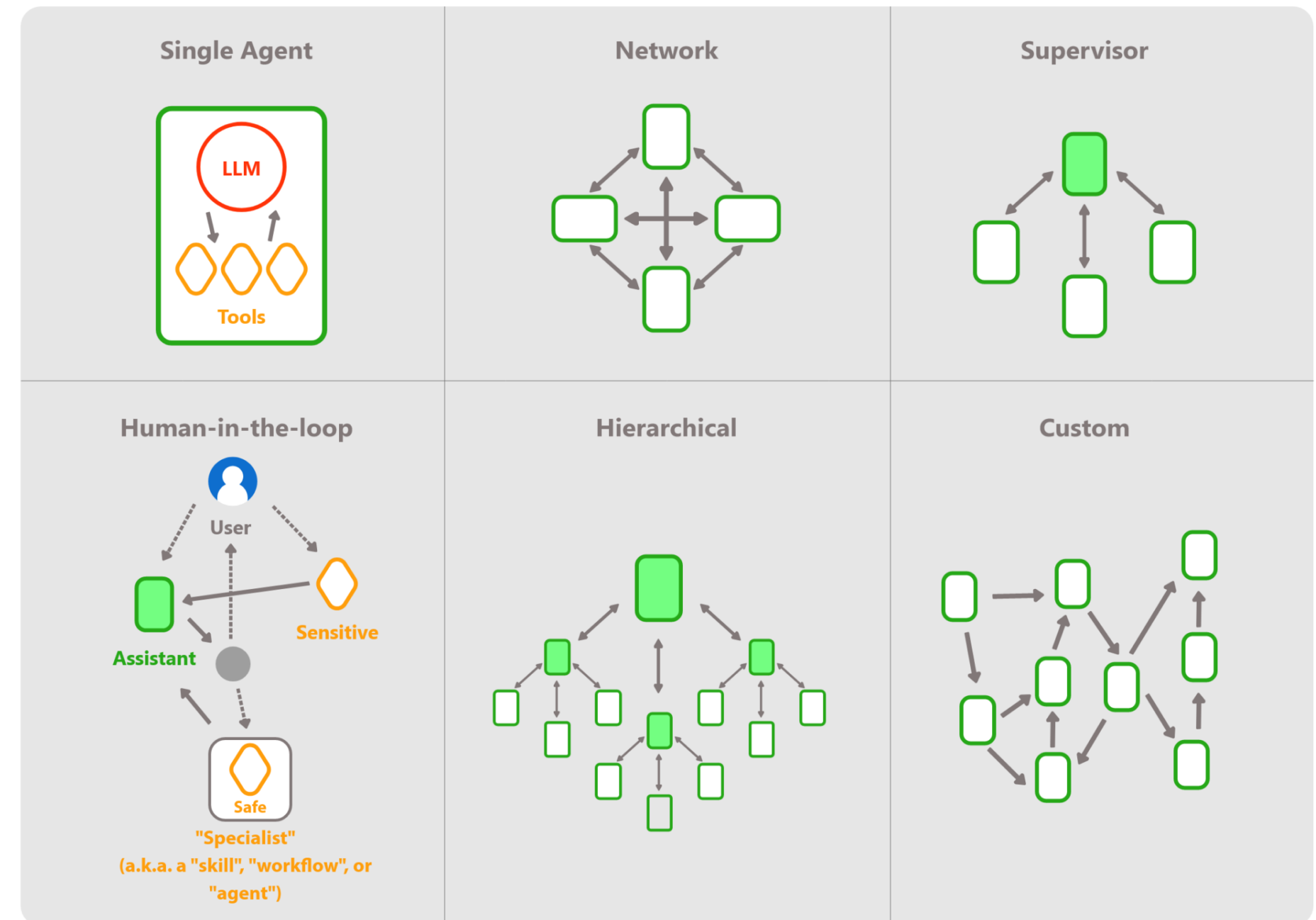
Document review, report drafting, tender/RFP analysis, and compliance checks — tasks that once took analysts days now run through AI agents that read, extract, and summarize at scale.

2. Agentic workflows over single-shot AI

Industry has moved from "ask a chatbot a question" to multi-step agents that plan, call tools (search, databases, APIs), and self-correct — e.g. multi-agent RAG pipelines for domain-specific research.

3. AI embedded in existing systems

Rather than standalone apps, AI is wired directly into ServiceNow, GitLab, HR systems, and internal portals via integrations — meeting users inside tools they already use.



How AI is Transforming Everyday Life



Speed of information

Hours-long research or paperwork now takes minutes — people get answers, drafts, and decisions in real time instead of waiting on manual review cycles.



Shift in daily work

Professionals spend less time on repetitive extraction/summarization and more on judgment, review, and decisions — the job becomes "supervise the agent," not "do the task."

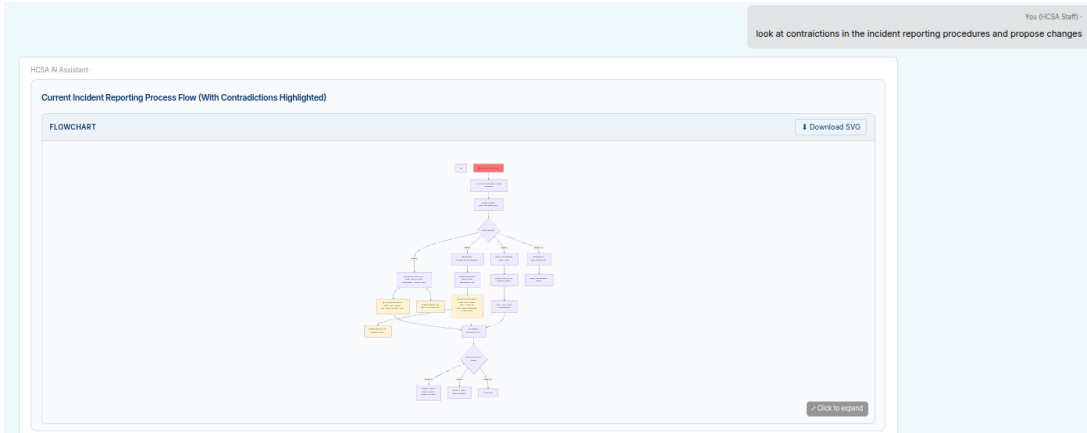


New expectations

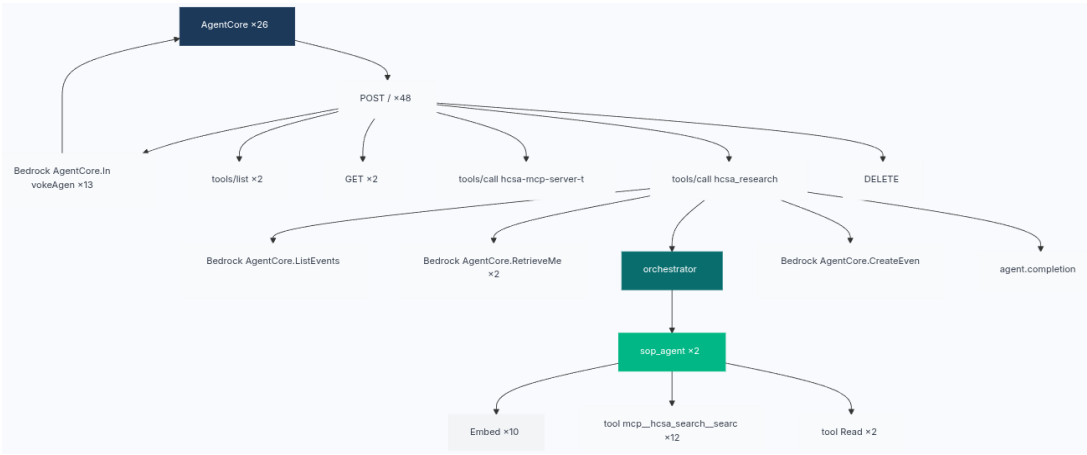
People now expect systems to explain themselves, respond conversationally, and act proactively — raising the bar for every digital service, not just "AI products."

Our Tech Stack

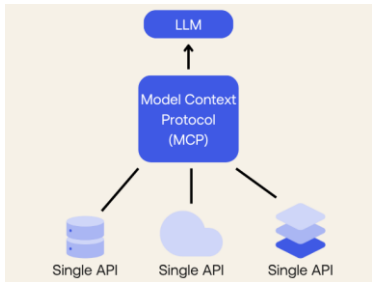
AWS Bedrock AgentCore



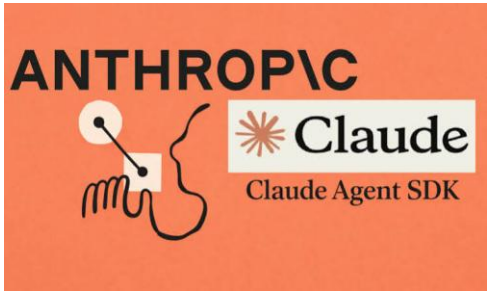
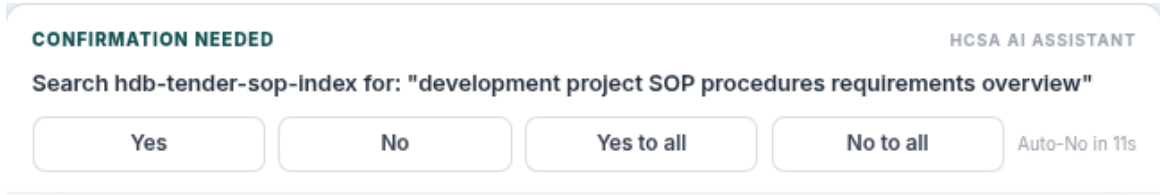
AG-UI: Agent-UI Protocol
Agentic UI. Exposing tools to agents for dynamic rendering



OTEL: OpenTelemetry
Traceability & Observability



MCP: Model Context Protocol
Standardizes how agents interact with external systems like APIs, databases, etc.



Agentic Harness: LangGraph, DeepAgents ClaudeAgentSDK



Meet the Speaker

Yu Kaiwen, Kevin

CEO
JoinAI



AI's Next Frontier Is the Physical World

The last wave changed how we create and process information. The next wave has to act in the real world.

THE LAST WAVE



Create and process information

Answers, drafts and summaries —
all of it inside software.



THE NEXT WAVE



Perceive, Predict and Act

Act in the real world.

WHERE IT LANDS



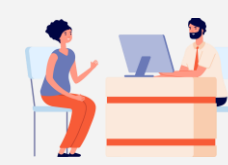
Manufacturing



Logistics



Healthcare



Services

...and every environment where intelligence has to move something real.

Embodied AI is Physical AI in Action

Physical AI needs a body. Software AI can stop at a recommendation — embodied AI has to close the loop.



01 · THE AMBITION

Physical Work

Intelligence that understands and changes the physical world.



02 · THE VALUE

Embodied AI

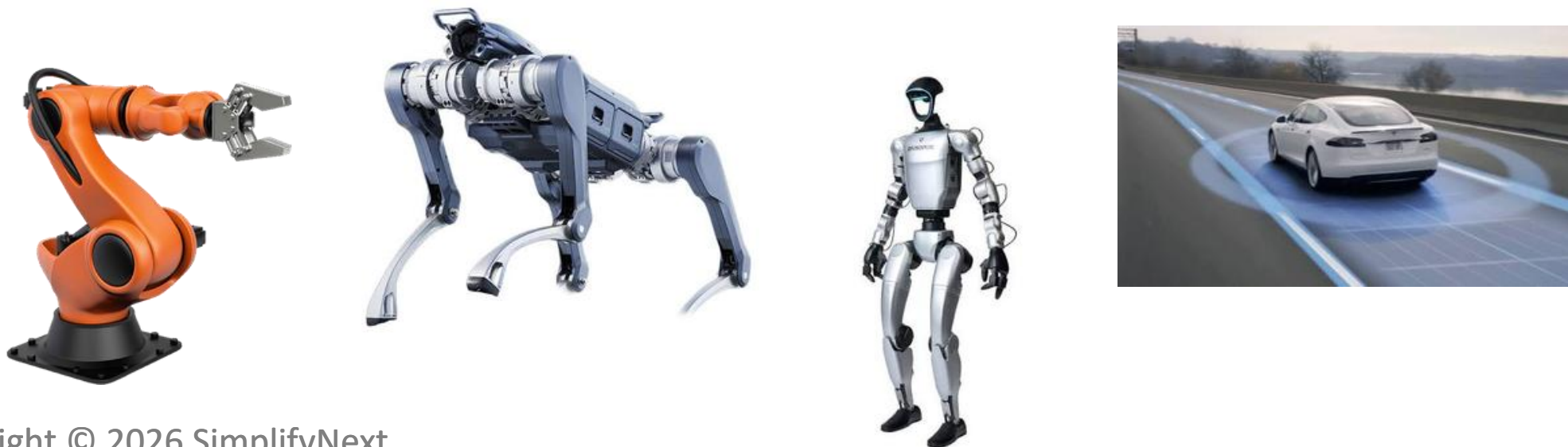
An agent with sensors, a body and the ability to act.



03 · THE REQUIREMENT

Closed Loop

Perceive, reason, act, and learn continuously.



The body doesn't have to be humanoid.

A robotic arm, a quadruped, a vehicle — any machine that senses and acts qualifies.

World Models Are Embodied AI's "LLM"

They predict how the physical world may change under different actions.



An LLM

predicts what comes next in language.



A world model

predicts how a physical state changes under an action.

THAT GIVES AN EMBODIED AGENT AN INTERNAL WORLD

Perceive

Fuse vision, touch, force and motion into a state the robot can reason about.



Predict

Simulate how the world may change under each possible action – before acting.



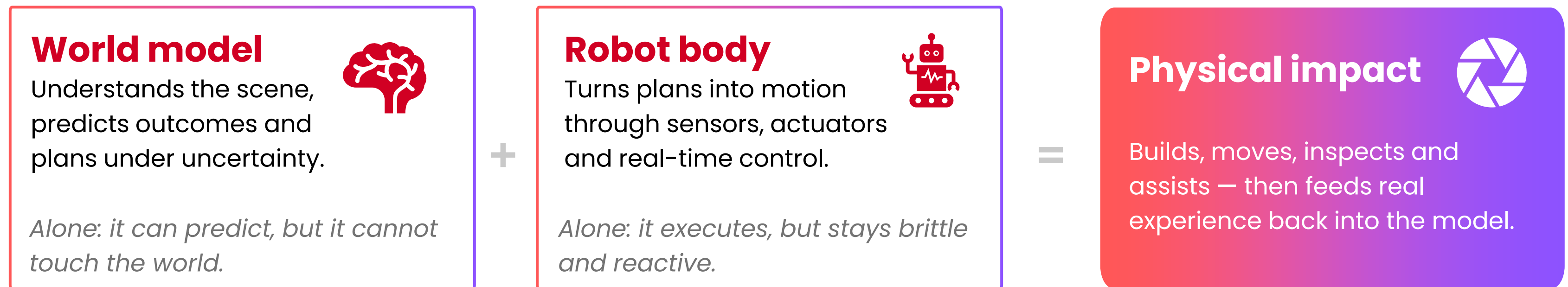
Plan

Choose the action that reaches the goal while respecting physics, safety and uncertainty.

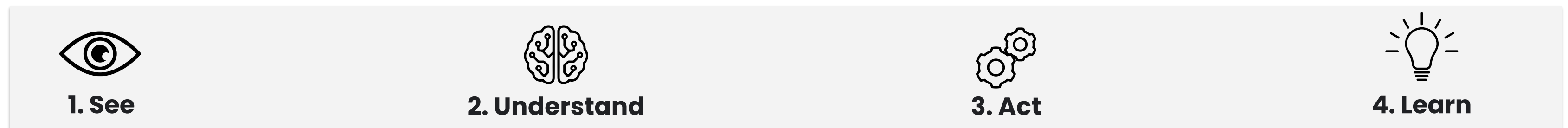
Perceive the state → predict the future → plan the action → act, then repeat.

World Models + Robots Bring AI Into the Physical World

World models provide foresight. Robots provide agency. Neither is enough on its own.



AND THE LOOP CLOSSES



Across manufacturing, logistics, infrastructure and human assistance.

Universities Are the Flywheel of Physical AI

JoinAi

Physical AI is a full-stack discipline: long-horizon research, shared infrastructure, skilled builders and real testbeds.



Talent

Students learn to build complete systems across data, models, simulation and hardware.



Research

Shared platforms turn new ideas into reproducible experiments and open benchmarks.



Ecosystem

Researchers, startups and industry connect, so progress compounds across cohorts.

...and every cohort feeds the next.

Universities give

Talent, research and new companies.



Industry returns

Real problems, real tools and deployment feedback.



This hackathon is that ecosystem in miniature.

JoinAI Builds Physical AI's Experience Layer

JoinAI

The bottleneck isn't more video — it's aligned experience.

ALIGNED EXPERIENCE MEANS FOUR THINGS, LINKED

What it saw

What it did

What changed

Did it
succeed

01 World-leading data infrastructure

Capture multimodal experience across industrial, service and home environments.

02 World-model-ready experience

Align state, action and outcome so models can learn dynamics, causality and control.

03 From data to deployment

Generate, augment and evaluate the data that moves agents past scripted demos.



For builders here: shorten the path from an idea to a model you can measure beyond one scripted demo.

Songying ORCA is Our Training Ground



The shared simulation platform for this hackathon — one place to build, train and evaluate before any hardware is involved.

01 Create scenes

Build the environment the task lives in.

02 Generate tasks

Turn scenes into repeatable objectives.

03 Train models

Learn behaviour against those tasks.

04 Evaluate behaviour

Score it before it touches hardware.



Iterate quickly

Test failures safely

Make results reproducible

Then carry the strongest behaviour into the real world.

Unitree Provides the Hardware + SDK

UNITREE 宇树

The point where a model stops being a model and becomes motion.

Motion

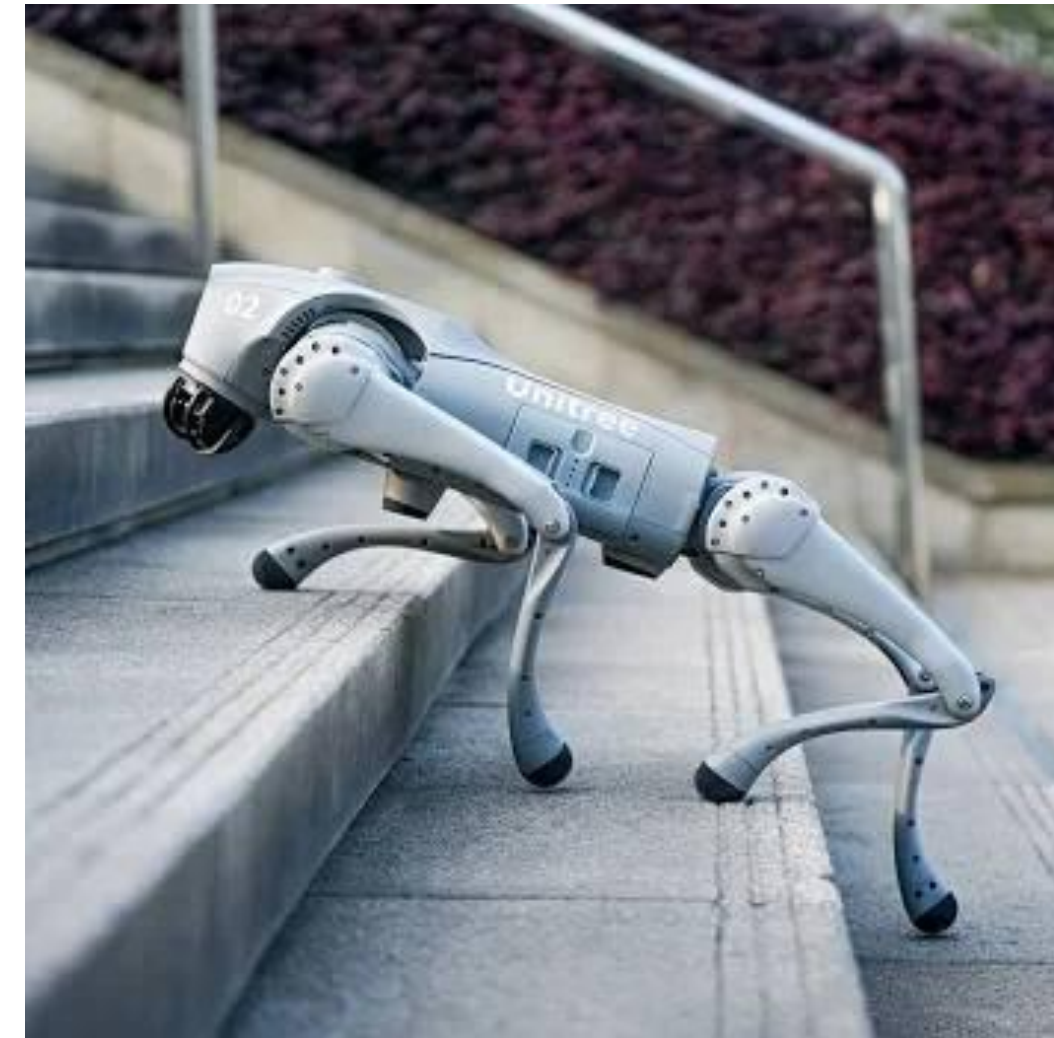
Plans leave the model and become actuator commands.

State feedback

The body reports what actually happened, not what was intended.

Real-world consequence

Success and failure are physical — you can't score them offline.



The loop is complete

JoinAI gives the experience · **ORCA** gives a world to train and test in · **Unitree** gives the body.

Build Something That Still Works When the World Changes

JoinAI

Experience

Aligned data worth learning from.




NaVILA: Legged Robot Vision-Language-Action Model for Navigation

+

SONGYING ORCA

World

A place to train, test and reproduce.




ORCA Lab 1.0
开发者版

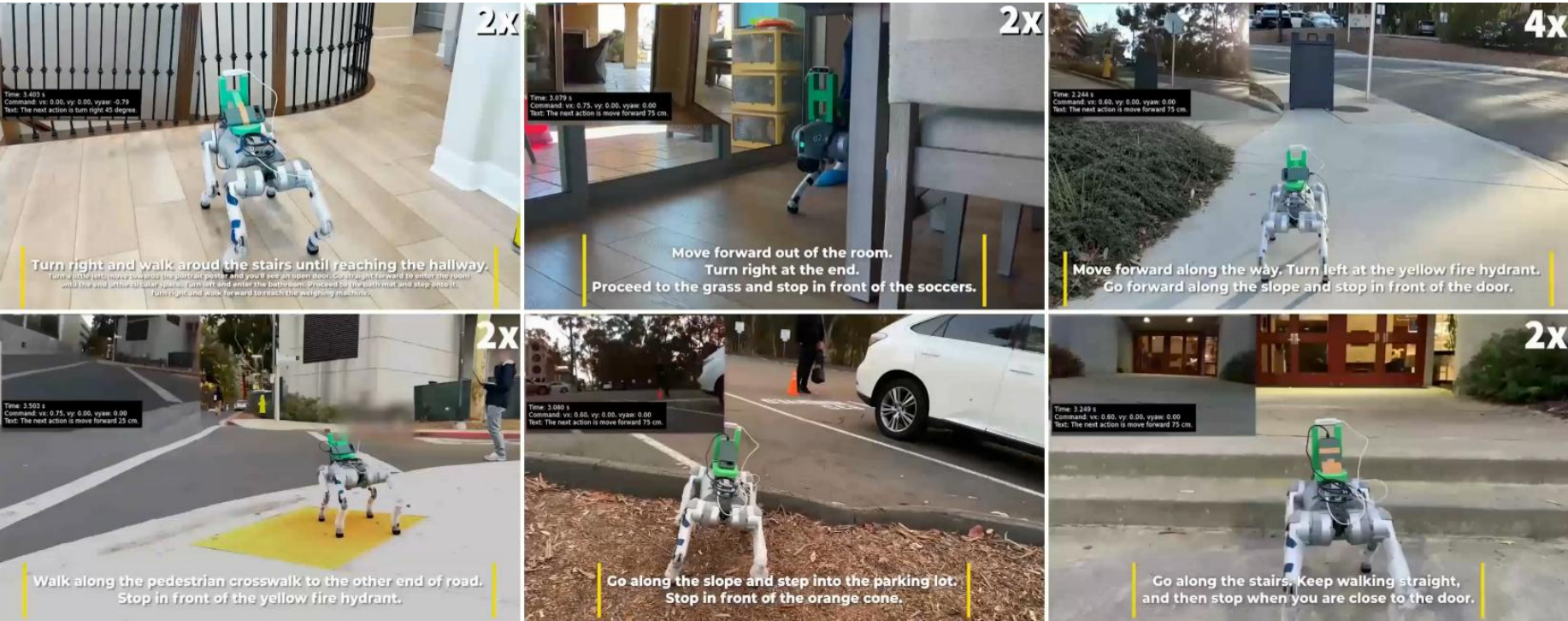
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UNITREE

Body

Where the plan meets physics.



Good luck — we're looking forward to what you build.

Training Sessions

We'll equip you with the essential skills to kickstart your Agentic AI journey. Our hands-on training sessions are designed to build your foundation in AI toolsets – so you're ready to create impactful solutions.

Session	Date & Time	Agenda
Introduction to LLMs & Agents Introduction to Orca Sim	17 August 630pm to 930pm Various Locations	Software AI track – How to call an LLM and building your first agent (1.5hrs) Physical AI track – Intro to Orca Sim (1.5hrs) <i>Attendees to sign up for free Groq accounts to support learning</i>
LangGraph and Beyond Navila in Orca Sim	18 August 630pm to 930pm Various Locations	Software AI track – Multi-Agent frameworks (1.5hrs) Physical AI track – Navila and running navila in orca sim (1.5hrs)

Training Sessions

We'll partner closely with participants throughout the hackathon – offering guidance to shape your ideas, choose the best development approach, and optimize your solution.

Session	Date & Time	Agenda
Solution Development Best Practices & Project Optimization	24 August 7pm to 9pm Various Locations	Best practices, code readability, project folder structure, presentation prep, Gen AI-powered documentation, project optimization, impact evaluation and open Q&A.
Intro to UniTree robot SDK Live demonstration Hackathon Submission	25 August 7pm to 9pm Various Locations	Physical AI track – Intro to UniTree robots and live demo of SportsClient (2hrs) Hackathon Admin (30 mins)

Meet the Speaker

Pearlyn Wong

Talent Acquisition Senior Manager
SimplifyNext

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Meet the Trainers



Arita Seah



Chee Wan Lee



Daryl Chia



Felicia Leow



Gerald Koh



Joon Sern Lee



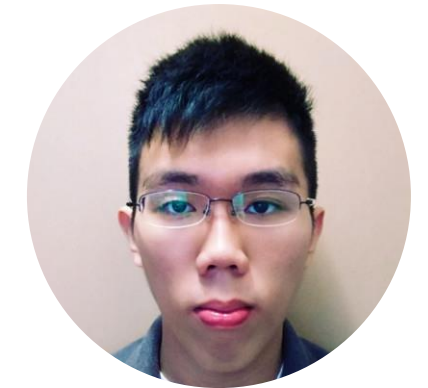
Mark Chew



Ricky Sutarius



Samuel Hoong



Ting Yew Koh



Thet Su Win



Wai Kit Tang



Wesley Tan



Xin Yu Li



Zhao Hui Ang



Zi Liang Chai

Meet the Trainers



EnZe Lee
UniTree

UNITREE 宇树



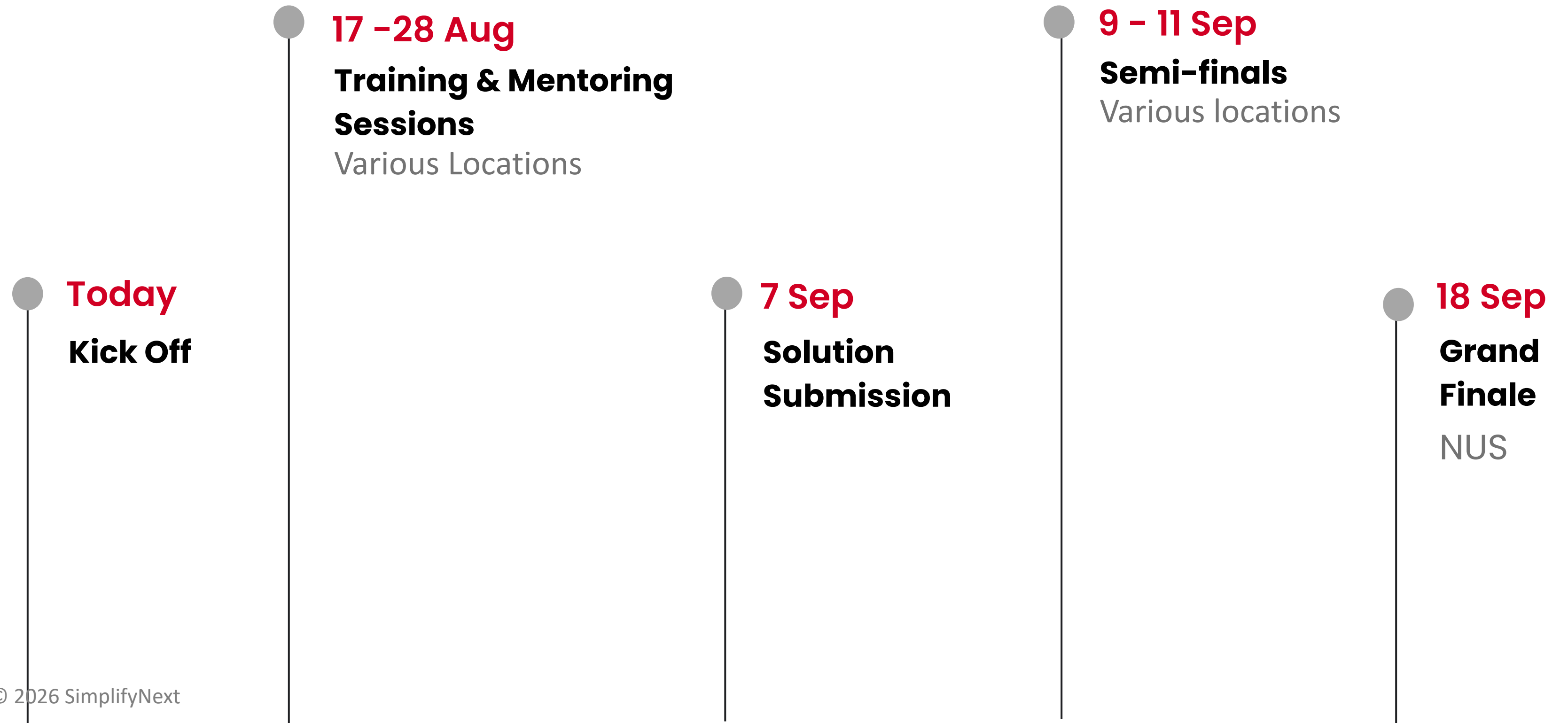
ZeMing Wang
SongYing



GuanZheng Liu
SongYing



Key Hackathon Dates



Prizes for all!



CHAMPION



1ST-RUNNER UP



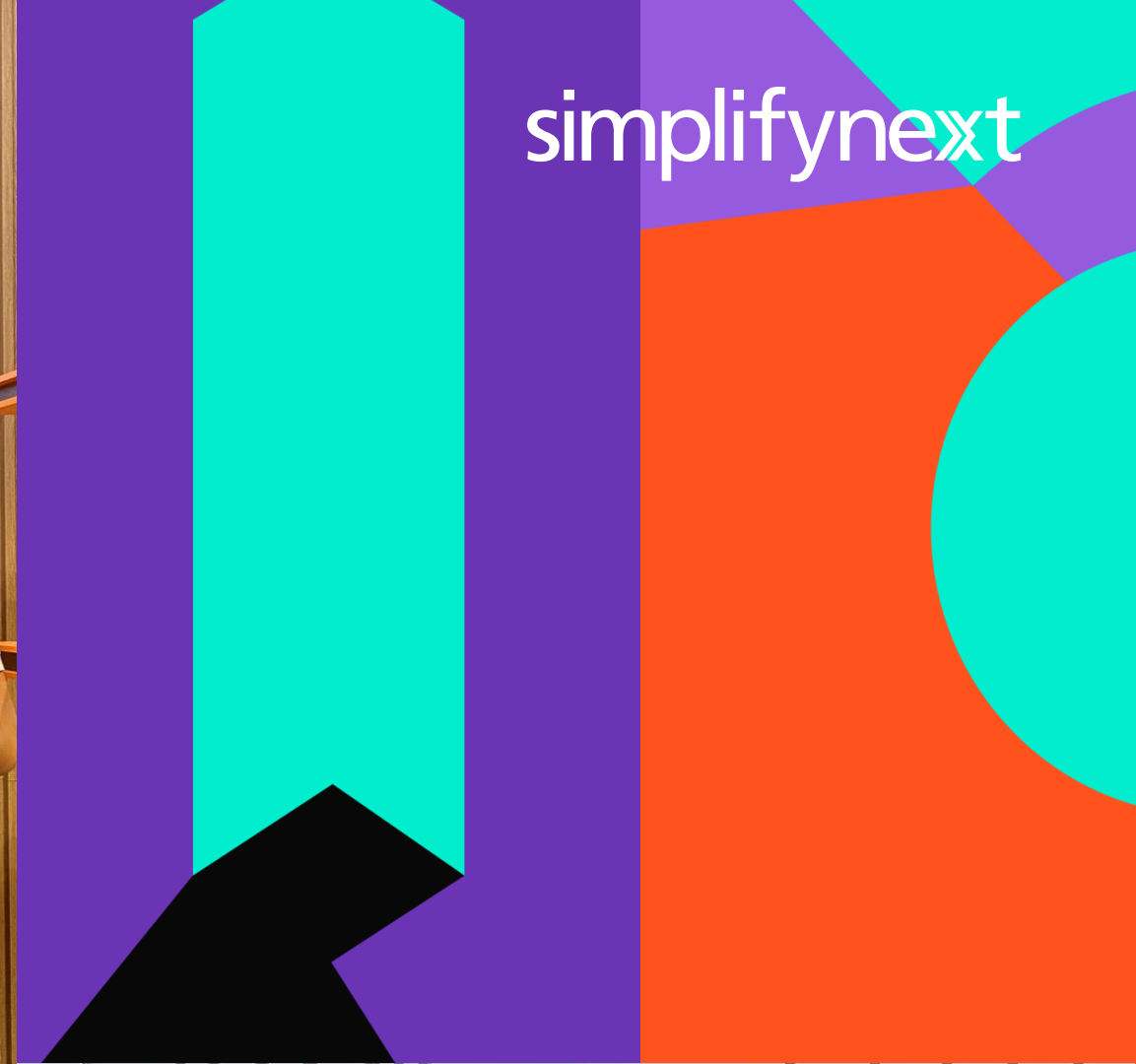
2ND-RUNNER UP

Honourable Mention

Tech Gadgets

Finalists

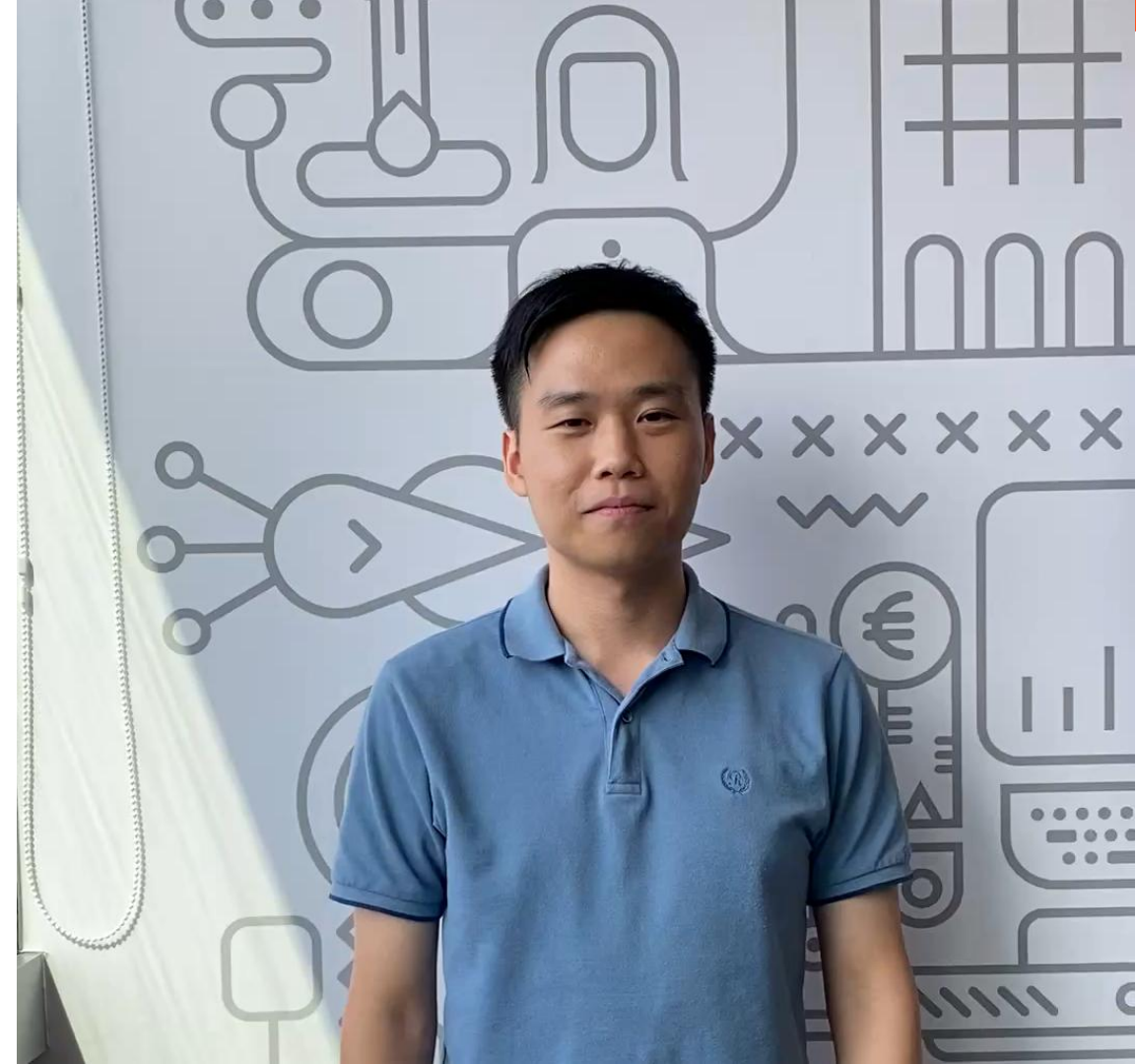
Exclusive Swags



From Hackathon to Hired

"I did the hackathon with the downtime I had while looking for a job. Instead it broadened my horizons and guided me to my first full-time job!"

*Lionel Wong,
2024 Hackathon winner | Technology Analyst, SimplifyNext*



Their Advice, Their Journey

“Start with the problem, not the solution! It's easy to get hyped building something technically cool and then hunting for a use case.

But it's important to find a real problem first and let that guide what you build. That's how you actually make a meaningful impact. ”

Aishwarya Goyal

2025 Hackathon Winner | Intern, SimplifyNext



“Honestly, massive technical growth! It pushed me to dive into building scalable, serverless backends and completely changed how I approach engineering.

I went from building things just for functional purposes to incorporating intuitive and secure features for the user.”

Kelly Ang

2025 Hackathon Winner | Intern, SimplifyNext